

How to Link the Static Library (.a) to a STM32CubeIDE Project

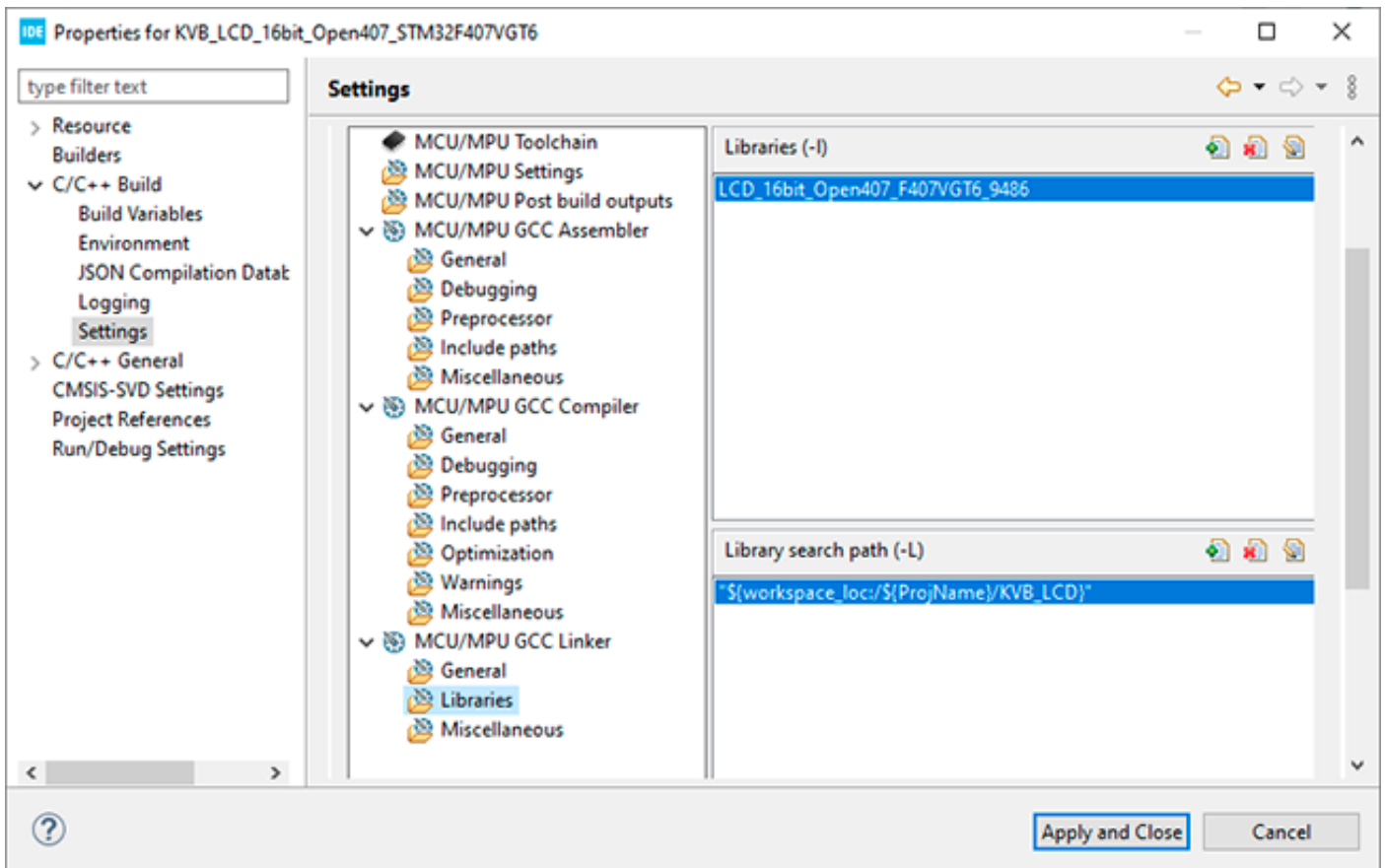
The LCD panel static graphic libraries (.a) are located within the **KVB_LCD** project folder, alongside the **GUI** and evaluation files (.c and .h).

File Structure Overview

- **KVB_LCD.h** – Main **GUI** (Graphical User Interface) header file.
- **KVB_TEST.h** – Header file containing two evaluation/presentation functions tailored for different screen resolutions.
- **KVB_TEST.c** – Source file implementing the two resolution-dependent evaluation functions.
- **KVB_TEXT.h** – Header file dedicated to the text variables used in the evaluation code.
- **KVB_TEXT.c** – Source file containing the text variable definitions for the evaluation code.
- **libLCD_16bit_Open407_F407VGT6_9325.a** – Static library file for the **ILI9325 LCD panel (320 x 240 resolution)**.
- **libLCD_16bit_Open407_F407VGT6_9341.a** – Static library file for the **ILI9341 LCD panel (320 x 240 resolution)**.
- **libLCD_16bit_Open407_F407VGT6_9481.a** – Static library file for the **ILI9481 LCD panel (480 x 320 resolution)**.
- **libLCD_16bit_Open407_F407VGT6_9486.a** – Static library file for the **ILI9486 LCD panel (480 x 320 resolution)**.
- **libLCD_16bit_Open407_F407VGT6_8009A.a** – Static library file for the **OTM8009A LCD panel (800 x 480 resolution)**.

Linker and GUI Configuration Steps

1. In the **Project Explorer** panel, right-click the root of your project folder and navigate to **Properties -> C/C++ Build -> Settings**, then open the **Tool Settings** tab.
2. Scroll down to the **MCU GCC Linker** section and click on the **Libraries** sub-item.
3. You will see two configuration windows: **Libraries (-l)** (the upper window) and **Library search path (-L)** (the lower window).
4. In the lower window (**Library search path -L**), add the path to the directory where your physical library file (e.g., **libLCD_16bit_Open407_F407VGT6_9486.a**) resides.
5. In the upper window (**Libraries (-l)**), add your library's name. **CRITICAL REQUIREMENT:** You must input the name strictly **WITHOUT** the lib prefix and **WITHOUT** the .a extension (for example, type exactly: **LCD_16bit_Open407_F407VGT6_9486**). *Note: Only one graphic library must be added to the project configuration at a time.*
6. Click **Apply and Close**.



7. Open the main GUI header file **KVB_LCD.h**, locate **line 120**, and uncomment *strictly* the line corresponding to your target **LCD** panel. This line contains the averaged timing configuration for the panel's **FSMC** bus. These values depend heavily on the specific hardware implementation of the panel, wire lengths, and ribbon cable traits.

⚠ CRITICAL HARDWARE NOTE: Due to the extremely high data transfer rates into the display's **GRAM**, if you are using a flat ribbon cable, it is **MANDATORY** to install a **22 pF** signal reflection dampening capacitor at the very end of the ribbon cable. Place it as close as physically possible to the panel's interface connector on the **WE (Write Enable)** signal line.

8. Finally, navigate to your project's **main.c** file and uncomment one of the two evaluation functions depending on your display's resolution: **lcd_Test480x320Landscape()** or **lcd_Test320x240Landscape()**.

During the compilation process, **STM32CubeIDE** will automatically fetch and link the pre-compiled binary code from your specified library file (e.g., **libLCD_16bit_Open407_F407VGT6_9486.a**).